Einstein's Equations from Substrate Closure

The Stress-Tensor Reidentification, Fierz–Pauli by Chart-Freedom Uniqueness, and the Bootstrap Completion of the VFD Gravity Chain

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Abstract

The VFD gravity chain previously stood at: Newtonian limit derived from the substrate boundary response [Smart, 2026a]; Schwarzschild compactness $\kappa = 2GM/(rc^2)$ conditional on four named hypotheses; and a Fierz–Pauli conjecture blocked by a documented obstacle — the substrate action is quadratic in the field F, hence apparently quartic in the candidate metric perturbation $h_{\mu\nu} \sim \langle \partial F, \partial F \rangle$, so no quadratic spin-2 action seemed derivable.

This paper closes the mechanism gap in the chain, at the effective level made precise in §13. The key step is a reidentification: the symmetric bilinear $\langle \partial_\mu F, \partial_\nu F \rangle$ is not the metric perturbation but (up to its trace part) the canonical Noether *stress-energy tensor* of the substrate field — the source of gravity, properly quadratic in matter. The metric perturbation is instead the coarse *collective* deformation field of the rendered scene; its effective action is quadratic in the Gaussian/large-N regime by a Wishart argument, independently of the microscopic action's order in F. Three substrate-derived inputs — a symmetric rank-2 collective field, second-order time, and invariance under chart redescription — then force the Fierz–Pauli action by the classical uniqueness theorem, which we re-derive numerically from scratch. Trace reversal, the weak-field factor of two, the full spatial metric components, and light deflection $4GM/(bc^2)$ become theorems rather than ansatz inputs; the conserved coupling and its universality (the equivalence principle) follow from chart-freedom consistency; and the unique nonlinear completion is the Einstein field equations with cosmological constant, via the classical Gupta–Kraichnan–Feynman–Deser bootstrap. The one constant the chain leaves free, Λ, is fixed independently by the programme's cosmology branch [Smart, 2026b].

Two bridges are included. Quantum: the envelope of a massive substrate mode obeys the free Schrödinger equation with $m = \hbar\omega_0/c^2$, and the inertial and gravitating masses of one simulated wavepacket agree quantitatively, making the equivalence principle a theorem of the one-field origin. Standard Model: the same uniqueness

mechanism one rank down forces the Maxwell action; the non-abelian gauge group content and the heavy sector remain open and are listed as such.

All 39 verification items pass, each with a genuine falsifiable null. The remaining gaps are rigor-grade, not mechanism-grade: the strict discrete-to-continuum limit (the Paper XL programme), the Gaussian-regime restriction, and the imported classical bootstrap theorems.

1 Introduction and positioning

The VFD programme derives physical structure from the 600-cell substrate $V_{600} \subset S^3 \subset \mathbb{H}$ and its closure dynamics. For gravity, the state of the chain before this paper was recorded in the kappa derivation notes [Smart, 2026a] and the continuum-targets programme paper [Smart, 2026d]:

- *Newtonian tier (closed).* The boundary response of a σ-paired point source gives $\Phi(r) = -GM/r$ and $\vec{g} = -GM/r^2\,\hat{r}$, with the Poisson equation $\nabla^2\Phi = 4\pi G\rho_m$ from a substrate scalar action.
- *Schwarzschild tier (conditional).* $\kappa(r) = 2GM/(rc^2)$ held only under four hypotheses: (H-wave) an emergent wave operator, (H-tensor) a rank-2 perturbation, (H-stress) a stress tensor from the σ-paired residual, and (H-trace) the trace-reversed coupling with its 16π normalisation.
- *Fierz–Pauli tier (blocked).* The direct construction failed on a documented obstacle: the natural substrate action is quadratic in the quaternionic field F, while the candidate metric $h_{\mu\nu} = \langle \partial_\mu F, \partial_\nu F\rangle$ is bilinear in F, so any action quadratic in h is quartic in F. Three obstacles were named; the first was load-bearing.

This paper closes the conditional tier and dissolves the blocked one. The structure of the argument is worth stating plainly, because it is not a continuation of the failed construction but a correction of its premise.

The reidentification. The bilinear $\langle \partial_\mu F, \partial_\nu F\rangle$ was being read as the metric perturbation. It is not. Completed with its trace term it is the canonical Noether stress-energy tensor of the substrate field (Theorem 1) — conserved on shell, positive in energy, exactly conserved tick-by-tick in the discrete dynamics on the actual 600-cell graph (Lemma 2). "Quadratic in F" is then exactly right: the *source* of a field equation is quadratic in the matter fields in every field theory. Obstacle 1 was a category error, and once the object is moved to the correct side of the field equation there is nothing left of it.

What gravity is instead. The metric perturbation is identified as the coarse, kernel-smoothed *collective* deformation of the rendered scene (Definition 5) — a constrained statistic of many substrate modes, in the same sense in which a phonon field is a collective coordinate of an atomic lattice. Its effective action is quadratic at leading order in the coarse limit regardless of the microscopic action's order (Proposition 7), and it transforms as a rank-2 tensor under the substrate symmetry (Proposition 6, verified on V_{600} itself).

Why the dynamics is forced. The rendered chart is observer bookkeeping; no substrate quantity depends on it. An infinitesimal redescription of the chart shifts the rendered metric components by exactly a linearised diffeomorphism (Proposition 8), so the effective action must be gauge invariant. The most general two-derivative quadratic action for a symmetric rank-2 field has six parameters; gauge invariance collapses the solution space to one dimension — the Fierz–Pauli action, with the mass terms killed and two physical polarisations (Theorem 10). The substrate does not construct the spin-2 dynamics; it supplies premises whose unique consistent consequence the dynamics is.

What then falls out. Varying the unique action yields the linearised Einstein tensor, so the trace-reversed field equation and its factor of two are theorems (Theorem 14); the full weak-field Schwarzschild metric, including the spatial components the earlier Nordström-style fallback could not produce, follows by the Green-function computation already on record; light bending comes out at $4GM/(bc^2)$, twice the Newtonian-corpuscle value, and we verify this by tracing rays through the numerically solved metric with a scalar-gravity control giving exactly half (§8). Gauge-invariance consistency forces the source to be the conserved stress tensor, and the one-substrate origin of all matter forces one coupling constant for everything — the equivalence principle as a structural fact (§7). The nonlinear completion to the full Einstein equations rides the classical bootstrap [Gupta, 1954, Kraichnan, 1955, Feynman et al., 1995, Deser, 1970, Boulware and Deser, 1975, Wald, 1986], imported and named as such (§9).

Honesty contract. Section 13 lists the named residuals. They are rigor-grade: the strict discrete-to-continuum limit (the Paper XL targets [Smart, 2026d]), the Gaussian-regime restriction on the effective-action argument, the imported bootstrap theorems, and the calibration status of G. None is of the missing-mechanism kind the previous obstacles were. The derivation stands at the level at which condensed-matter physics derives sound from atoms: effective, quantitative, numerically verified, and not yet a continuum theorem in the strict mathematical sense.

2 Substrate inputs

All inputs are previously derived; none is new to this paper.

(I1) *Wave dynamics.* Second-order time is forced by the inner clock reversal: elements $g \in 2\mathbf{I}$ with $g\tau g^{-1} = \tau^{-1}$ exist, and equivariance of the residual law under the induced involution forbids first-order ∂_t terms [Smart, 2026c]. The spatial generator satisfies $L_\partial/a^2 \to -\nabla^2$ in the long-wavelength limit, with the exact dispersion relation on V_{600} as witness. Together: $\Box = c^{-2}\partial_t^2 - \nabla^2$ with $c = a/\tau$. The remaining second-difference formality is closed in §10.1.

(I2) *Arena and chart.* The rendered arena is the intrinsic S^3 with spectral dimension 3; the canonical kernel is $\pi_r = (I + r^2 L)^{-1}$; the anchor chart is the quaternion-exponential normal chart with frame $\{v_0 i, v_0 j, v_0 k\}$ [Smart, 2026c]. The chart is explicitly non-canonical.

(I3) *Newtonian tier.* $\nabla^2\Phi = 4\pi G\rho_m$, $\Phi = -GM/r$ [Smart, 2026a].

(I4) *Quaternionic substrate field.* $F : V_{600} \times \mathbb{Z} \to \mathbb{H}$ with tangent-frame derivatives $\partial_\mu F$ and quadratic action $S_F = \int[-\frac{\alpha}{2}\langle \partial_\mu F, \partial^\mu F\rangle + \langle F, J\rangle]$, equation of motion $\alpha\,\Box F = -J$ [Smart, 2026a].

(I5) *Response family.* $C_\varphi = L_\partial + \varphi^{-2} I$: the $\varphi^{-1} \to 0$ end is the Newton regime; finite φ^{-1} is the massive (Yukawa) regime [Smart, 2026a]. Used for the quantum bridge in §11.

Signature is $(-,+,+,+)$ throughout; $\eta = \mathrm{diag}(-1,1,1,1)$.

3 The bilinear is the stress tensor

Theorem 1 (Canonical stress tensor; discharges (H-stress))**.** *For the substrate field of (I4) with $\alpha > 0$, on shell in the source-free sector ($J = 0$), the canonical Noether current of spacetime-translation invariance is*

$$\Theta_{\mu\nu} \;=\; \alpha\left[\langle \partial_\mu F, \partial_\nu F\rangle \;-\; \tfrac{1}{2}\,\eta_{\mu\nu}\,\langle \partial_\rho F, \partial^\rho F\rangle\right], \tag{1}$$

which is (i) symmetric; (ii) conserved on shell, $\partial^\mu \Theta_{\mu\nu} = \alpha\langle \Box F, \partial_\nu F\rangle = 0$ when $\Box F = 0$; and (iii) of non-negative energy density, $\Theta_{00} = \frac{\alpha}{2}\big(\langle \partial_0 F, \partial_0 F\rangle + \langle \partial_i F, \partial_i F\rangle\big) \geq 0$.

Proof. Direct computation: $\partial^\mu \Theta_{\mu\nu} = \alpha\big[\langle \Box F, \partial_\nu F\rangle + \langle \partial_\mu F, \partial^\mu \partial_\nu F\rangle - \langle \partial_\rho F, \partial_\nu \partial^\rho F\rangle\big]$, and the last two terms cancel. Symmetry is manifest in (1); positivity of Θ_{00} is a sum of squares. □

On the discrete substrate the time-direction statements are exact, not approximate.

Lemma 2 (Exact discrete conservation on V_{600})**.** *Let $F_{t+1} = 2F_t - F_{t-1} - \varepsilon^2 L F_t$ be the leapfrog substrate dynamics on the 600-cell graph Laplacian L. Then:*

(a) the discrete energy $E_{t+1/2} = \frac{1}{2}\|(F_{t+1} - F_t)/\tau\|^2 + \frac{c^2}{2a^2}\langle F_t, L F_{t+1}\rangle$ is conserved tick-by-tick exactly (sim T5: relative drift $< 10^{-10}$ over 2000 ticks);

(b) for the semi-discrete dynamics $\ddot{F} = -(c^2/a^2) L F$, the local energy $e_v = \frac{1}{2}\dot{F}_v^2 + \frac{c^2}{4a^2}\sum_{w\sim v}(F_v - F_w)^2$ and edge flux $J_{vw} = \frac{c^2}{2a^2}(F_v - F_w)(\dot{F}_v + \dot{F}_w)$ satisfy the exact continuity equation $\dot{e}_v + \sum_{w\sim v} J_{vw} = 0$ at every vertex (sim T6: machine precision).

Proof. (b) is a two-line computation from $\ddot{F}_v = -(c^2/a^2)\sum_{w\sim v}(F_v - F_w)$, with squares read as quaternion-component sums ($\dot{F}_v^2 := \langle \dot{F}_v, \dot{F}_v\rangle$, etc.); the flux J_{vw} is antisymmetric in $v \leftrightarrow w$, so the law is a genuine local conservation law on the graph. (a) is the standard Störmer–Verlet invariant for symmetric L. □

Lemma 3 (Dust limit)**.** *For a slowly modulated carrier mode of the massive equation $(\Box - \mu^2)F = 0$ with envelope/carrier scale ratio ϵ, the period-averaged components satisfy $\langle \Theta_{11}\rangle / \langle \Theta_{00}\rangle = k^2/\omega^2 = O(\epsilon^2)$ (verified computationally to 10^{-6} at two values of k, item T7c, for the massive stress tensor including its μ^2 term; stated as a verified computation, not re-derived here). Matter wavepackets therefore source gravity as static dust at leading order.*

Remark 4 (Obstacle 1 dissolved). The previous diagnosis — "action quadratic in F is quartic in h" — assumed $h \sim \langle \partial F, \partial F\rangle$. With the bilinear identified as $\Theta_{\mu\nu}$, quadratic-in-F means quadratic-in-matter, which is what a source must be. The question becomes: where does the metric live? That is §4.

4 The collective metric field

Definition 5 (Rendered deformation field)**.** Let δF be the substrate fluctuation about the homogeneous closure background and π_r the canonical kernel of (I2). The rendered deformation field at chart point x is the coarse-grained, background-subtracted second moment

$$h_{\mu\nu}(x) \;\propto\; \pi_r[\langle \partial_\mu \delta F \, \partial_\nu \delta F \rangle]\,(x) \;-\; (\text{homogeneous background}), \tag{2}$$

a symmetric rank-2 field with 10 components — a constrained statistic of many substrate modes per kernel cell, not a pointwise function of F.

Proposition 6 (Tensoriality)**.** *Under the substrate symmetry $2\mathbf{I} \times 2\mathbf{I}$ (left/right icosian multiplication), the frame components h_{ab} transform as a rank-2 tensor: left multiplication transports the right-quaternionic frame trivially, $g(v\,i) = (gv)\,i$; right multiplication by g^{-1} rotates the frame by the adjoint action $q \mapsto gqg^{-1}$, so $h \mapsto R_g^\top h R_g$ with $R_g = \mathrm{Ad}(g) \in SO(3)$.*

Proof. Associativity of quaternion multiplication and orthogonality of unit-quaternion translation. Verified to machine precision on the actual 600-cell with random fields, both actions (sims T14a–b); the kernel π_r commutes with the induced vertex permutations (T14c), so the coarse field transforms identically. □

Proposition 7 (Quadratic effective action)**.** *Let h be the sample second moment of N independent substrate modes per kernel cell in the Gaussian (linear-response) regime, with background Σ_0. The effective action (constrained log-density) is*

$$\Gamma[\Sigma_0 + \delta h] \;=\; \frac{N}{4}\,\mathrm{tr}\big[(\Sigma_0^{-1}\delta h)^2\big] + O\big(N\,\delta h^3\big)\,, \tag{3}$$

and since equilibrium fluctuations scale as $\delta h \sim N^{-1/2}$, the cubic term is $O(N^{-1/2})$ relative to the quadratic. In the coarse limit the effective action of the collective field is quadratic.

Proof. For Gaussian modes the sample covariance is Wishart [Wishart, 1928]; the log-density is N times an analytic function with a nondegenerate quadratic minimum at Σ_0. Sims T15a–c verify the quadratic coefficient (ratio 1.00001), the $N^{-1/2}$ decay of the cubic-to-quadratic ratio at the fluctuation scale, and the empirical $N^{-1/2}$ scaling of sample-second-moment fluctuations. □

This is the standard emergent-field mechanism — it is how phonons acquire a quadratic action from an anharmonic microscopic crystal — and it makes the order of the *microscopic* action in F irrelevant to the order of the *effective* action in h. It fixes the potential (zero-derivative) structure of the effective action; that the effective action is additionally local with at most two derivatives is the standard effective-field-theory assumption, stated here explicitly. Which two-derivative action then appears is forced next.

5 Gauge invariance from chart freedom

Proposition 8 (Chart redescription acts as linearised diffeomorphism)**.** *The anchor chart of (I2) is observer bookkeeping; no substrate quantity depends on it. An infinitesimal redescription $x \mapsto x + \xi(x)$ changes the components of the rendered metric $g = \eta + h$ by the Lie derivative; at linear order*

$$h_{\mu\nu} \;\longmapsto\; h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu + O(\xi\,\partial h,\; \xi^2). \tag{4}$$

Chart-independence of the substrate therefore requires the effective action $\Gamma[h]$ to be invariant under this shift.

Proof. The transformation law is the textbook pullback; sim T8 verifies it numerically by exact re-charting of a metric built from explicit Fourier modes, with the residual converging at $O(\xi^2)$ (ratio 4.00 under halving) and constituting a vanishing fraction of the shift. □

Remark 9. Two honesty notes. (i) The substrate has a preferred *frame* (the tick direction u^μ); a frame is a state of motion, not a chart restriction, and does not reduce the redescription freedom — the situation is the same as in cosmology, where the CMB frame coexists with diffeomorphism invariance. (ii) The discrete lattice has no continuous diffeomorphisms; the statement lives at the level of the rendered (kernel-smoothed) fields, where the chart is literally an arbitrary choice. The local-frame non-canonicity previously flagged as a worry in [Smart, 2026a] *is* the gauge freedom.

6 Fierz–Pauli by uniqueness

Theorem 10 (Uniqueness of the quadratic action)**.** *The most general local Lorentz-invariant action for a symmetric rank-2 field with at most two derivatives is, modulo total derivatives, the 6-parameter family*

$$S[h] = \int d^4x \Big[a_1 \partial_\rho h_{\mu\nu} \partial^\rho h^{\mu\nu} + a_2 \partial_\mu h^{\mu\nu} \partial^\rho h_{\rho\nu} + a_3 \partial_\mu h^{\mu\nu} \partial_\nu h + a_4 \partial_\mu h\, \partial^\mu h + b_1 h_{\mu\nu} h^{\mu\nu} + b_2 h^2\Big]. \tag{5}$$

Invariance under $h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$ for all ξ forces $b_1 = b_2 = 0$ and $(a_1, a_2, a_3, a_4) \propto (-\frac{1}{2}, 1, -1, \frac{1}{2})$: the solution space is exactly one-dimensional and is the Fierz–Pauli action [Fierz and Pauli, 1939]. The graviton is massless and carries exactly two physical polarisations.

Proof. Finite linear algebra in momentum space. Writing the action as a quadratic form $Q(h, h; k)$ and imposing $Q(h, \delta_\xi h; k) = 0$ for all (h, ξ, k) yields a finite-dimensional linear system on the six coefficients; the sampled system attains maximal rank, so the SVD is a rank certificate, not a statistical estimate: null space dimension 1, with the stated ratios to 10^{-9} and the mass terms vanishing (item T1). At null momentum the kinetic operator has a 6-dimensional kernel containing the 4-dimensional gauge orbit, leaving $10 - 4 - 4 = 2$ physical polarisations by the kernel/orbit dimension count (item T2; the standard constraint analysis in harmonic gauge gives the same count, Hinterbichler, 2012). The result is the classical Fierz–Pauli uniqueness theorem (see Hinterbichler, 2012 for a modern review); the simulation re-derives it from scratch so that no step of the chain assumes it. □

Corollary 11. *The variation of the Fierz–Pauli action is the linearised Einstein tensor: $\delta S_{FP}/\delta h^{\mu\nu} \propto G^{\rm lin}_{\mu\nu}[h]$, with one proportionality constant across random fields and momenta (sim T3a, residual $< 10^{-15}$).*

7 Coupling, conservation, universality

Proposition 12 (Conservation forced; coupling consistent)**.** *With matter coupling $S_{\rm int} = \lambda \int h_{\mu\nu} \Theta^{\mu\nu}$, gauge invariance of the total action requires $\int \xi_\nu\, \partial_\mu \Theta^{\mu\nu} = 0$ for all ξ: the* *source must be a conserved symmetric tensor, which $\Theta_{\mu\nu}$ is by Theorem 1. Conversely, the conserved symmetric rank-2 current of the substrate matter sector is unique up to improvement terms, so the coupling is unique up to normalisation [Weinberg, 1964].*

Proposition 13 (Universality; equivalence principle)**.** *All foreground matter consists of σ-paired modes of the one substrate field F, rendered through the same chart and kernel. There is therefore a single coupling λ for all matter: no per-species gravitational charge exists because there is only one substrate. A quantitative check is given in Theorem 21.*

8 Trace reversal, Schwarzschild, light bending

Theorem 14 (Trace-reversed field equation; discharges (H-trace))**.** *The stationary point of $S_{FP} + S_{\text{int}}$ in harmonic gauge is*

$$\Box\bar{h}_{\mu\nu} \;=\; -\frac{16\pi G}{c^4}\,\Theta_{\mu\nu}, \qquad \bar{h}_{\mu\nu} = h_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}h, \tag{6}$$

with the normalisation fixed by matching the Newtonian tier (I3). The trace reversal is generated by the variation of the kinetic term (Corollary 11); it is kinematics of the unique action, not a property of the matter coupling. For static dust the solution is

$$h_{00} = h_{11} = h_{22} = h_{33} = \frac{2GM}{rc^2}, \qquad h_{0i} = 0, \tag{7}$$

i.e. the full weak-field Schwarzschild metric in isotropic gauge, including the spatial components.

Proof. Corollary 11 plus the standard Green-function computation of [Smart, 2026a] §6, now unconditional. Sim T3 solves (6) on a 48^3 grid with a Gaussian dust blob: isotropy and the factor 2 hold exactly (machine precision, being algebraic identities of the trace reversal), the radial profile fits $2GM/r$ to 3%, and — the strong check — the harmonic-gauge solution satisfies the full gauge-unfixed linearised Einstein equation mode by mode with a single coupling constant (residual $< 10^{-15}$). ☐

Corollary 15 (Schwarzschild compactness, no remaining named hypotheses)**.** $\kappa(r) = 2GM/(rc^2)$ *holds with no remaining named physical hypotheses — what remains is the calibration of G and the rigor-grade residuals of §13. The earlier Nordström-style fallback, which produced only g_{00}, is retired.*

Corollary 16 (Light bending; the discriminator)**.** *Null rays in the derived metric deflect by $\alpha = 4GM/(bc^2)$, twice the Newtonian-corpuscle value, because $h_{ij} \neq 0$. Sim T4 traces rays through the numerically solved metric of T3 and recovers $4GM/(bc^2)$ to 5%; the h_{00}-only scalar control gives the ratio* 2.0000.

9 Nonlinear completion: the Einstein equations

Theorem 17 (Conditional on the classical bootstrap)**.** *A massless spin-2 field with gauge-invariant quadratic action, coupled universally to the total stress-energy including its own, has the Einstein–Hilbert action as its unique consistent nonlinear completion; hence*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} \;=\; \frac{8\pi G}{c^4}\,T_{\mu\nu}, \tag{8}$$

with Λ the single undetermined constant. This is the Gupta–Kraichnan–Feynman–Deser bootstrap [Gupta, 1954, Kraichnan, 1955, Feynman et al., 1995, Deser, 1970], closing in one step in first-order form [Deser, 1970], with uniqueness per [Wald, 1986] and the massive alternative excluded by [Boulware and Deser, 1975]. The substrate supplies all premises: massless spin-2 (§4–§6), universal conserved coupling (§7), and the self-coupling requirement ($\Theta_{00} \geq 0$ applies to the h-sector's own collective energy on the same one-substrate footing).

Remark 18 (The cosmological constant). The bootstrap permits Λ and cannot fix it. The programme fixes it elsewhere: the σ-paired complement determination of the published cosmology release (`hypersphere-cosmology` v1.1.0, `github.com/vfd-org/hypersphere-cosmology`) reproduces $\Omega_\Lambda = 0.6844$ (-0.083% against Planck), while the in-tree cascade route (Paper L [Smart, 2026b]) gives the cruder $\Omega_\Lambda = 2/3$ at 2.7%. The one constant the gravity chain leaves free is the one the cosmology branch determines, by an independent route — a consilience check, not a fit.

Remark 19 (Relation to the D_4/cascade route). Paper XL [Smart, 2026d] pursues the same target geometrically, by continuum limit of the 24-cell rung (targets T1–T4). The present paper closes the field-equation question at effective level first; Paper XL remains the home of the rigorous continuum-limit programme, i.e. of residual (R-cont) below. The routes are complementary.

10 Consequences

10.1 (H-wave), final formality

$\big(F_{t+1} - 2F_t + F_{t-1}\big)/\tau^2 = \partial_t^2 F + \frac{\tau^2}{12}\partial_t^4 F + O(\tau^4)$; combined with $L/a^2 \to -\nabla^2$ this gives $\Box$ with $c = a/\tau$. Sim T10 verifies both the Taylor step and, on the 600-cell itself, that the leapfrog eigenmode frequency converges to the continuum value at $O(\tau^2)$. (H-wave) is fully discharged.

10.2 Gravitational waves

In transverse-traceless gauge the field equation gives $\Box h_{\mu\nu}^{TT} = 0$: two polarisations (Theorem 10), dispersion $\omega = c|k|$ exactly (sim T9: h_+ and $h_\times$ solve the equation on the light cone and fail it off), and — because gravity and matter ride the same substrate wave operator (I1) — the gravitational-wave speed equals the light/matter speed identically, consistent with the GW170817 bound $|c_{GW}/c - 1| < 10^{-15}$ [Abbott et al., 2017] as a structural fact rather than a coincidence.

10.3 Time dilation and geodesics

$g_{00} = -(1 - 2GM/(rc^2))$ is Corollary 15; the inner-clock rate at depth Φ scales as $\sqrt{-g_{00}}$, with the time direction the real-quaternion clock axis. Conservation $\partial_\mu T^{\mu\nu} = 0$ in the curved background implies geodesic wavepacket motion at leading order by the standard argument [Geroch and Jang, 1975]; the null-ray case is exercised directly by sim T4.

11 The quantum bridge

Proposition 20 (Envelope → Schrödinger)**.** *A substrate mode at the massive end of the response family (I5) obeys* $(\Box - \mu^2)F = 0$*, i.e.* $\partial_t^2 F = c^2(\nabla^2 - \mu^2)F$*, with* $\mu = \varphi^{-1}$ *(consistent with the* $\Box$ *convention of (I1)). Writing* $F = \mathrm{Re}[\psi\, e^{-i\omega_0 t}\hat{q}]$ *with rest carrier* $\omega_0 = c\mu$ *and slowly varying envelope* ψ*,*

$$i\,\partial_t\psi = -\frac{c^2}{2\omega_0}\nabla^2\psi + O(\epsilon^2), \qquad \epsilon = \frac{c\,k_{\mathrm{env}}}{\omega_0}, \tag{9}$$

the free Schrödinger equation with $m = \hbar\omega_0/c^2$*. Sim T11 evolves the exact Klein–Gordon field spectrally and verifies the envelope error is* $O(\epsilon^2)$ *(ratio* 3.97 *under halving of* ϵ*). This complements the Nelson-pairing route of Paper XVIII [Smart, 2026e], which handles the interacting/equilibrium structure; the envelope route is the direct free-sector reduction.*

Theorem 21 (Equivalence principle, quantitative)**.** *For one wavepacket of the substrate field: the inertial mass read from its kinematics (momentum per quantum over group velocity) and the gravitating mass read from its stress tensor (energy per quantum, the quantity that sources (6)) agree identically, both being moments of the same field. Sim T12 measures both on one simulated packet:* $m_{\mathrm{grav}} = 8.0403$*,* $m_{\mathrm{inert}} = 8.0420$ *(carrier* $\mu = 8$*, deviations* $O(\epsilon^2)$ *as predicted).*

Remark 22 (Resolution bridge). The quantum and gravitational descriptions sample one substrate at two grains: the coarse-rung eigenfunctions are the fine-rung eigenfunctions sampled on shared vertices (the 24-cell inside the 600-cell; Smart, 2026d). Sections 9 and 11 are the two grains' effective laws.

12 The Standard-Model bridge: forced form, open content

Proposition 23 (Maxwell forced)**.** *The uniqueness scan of Theorem 10 applied to a rank-1 collective field* A_μ *with chart-phase redundancy* $A_\mu \to A_\mu + \partial_\mu\chi$ *collapses the general 3-parameter quadratic action to the 1-dimensional null space* $(1, -1, 0)$ *— the Maxwell action* $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ *— with two photon polarisations and conserved charge forced exactly as in Proposition 12 (sims T13a–c).*

The form of both long-range force laws is therefore forced by one theorem applied at rank 1 and rank 2. What the substrate has *not* supplied for the Standard Model, stated plainly: the non-abelian gauge group content (why $SU(3) \times SU(2) \times U(1)$); the heavy sector (W/Z/Higgs, heavy quarks) and neutrino placement. The existing contact points are the light-sector mass-ladder placements, the b-anomaly response-family shape (the same C_φ family whose two ends are Newton and Yukawa), and the conditional α-chain. This section is the honest boundary of the bridge.

13 Named residuals

(R-cont) *Continuum-limit rigor.* The chain operates at effective-field level, with the exact dispersion relation and $O(\tau^2)/O(a^2)$ convergence as witnesses. The strict discrete-to-continuum theorem is the Paper XL programme [Smart, 2026d]. This is now the only mathematically deep gap in the gravity chain.

(R-Gauss) Proposition 7 covers the Gaussian regime; strongly-coupled substrate regimes are not covered (physically, the deep-nonlinear region is taken over by the bootstrap completion, but the substrate-side statement is Gaussian).

(R-boot) Theorem 17 cites the bootstrap rather than re-deriving it. The cited results are mainstream and unconditional, but imported.

(R-G) G remains the substrate-to-physical mass calibration of [Smart, 2026a], Definition 4.2.

(R-Λ) Λ is not fixed by this chain; it is fixed independently by [Smart, 2026b] (Remark 18).

None of these is of the missing-mechanism kind that the previous Obstacles 1–3 were.

Update (second pass, June 2026). A dedicated residual-closure pass (`docs/residual-closure-de` sim suite `scripts/verify_residual_closure.py`, 24/24 PASS) has since upgraded this list:

- **(R-boot) closed in-house.** The first-order one-step completion (the $\eta^{\mu\nu} \to \eta^{\mu\nu} + h^{\mu\nu}$ substitution in the $\Gamma\Gamma$ term, closing in a single step) is now derived in the programme's own pages, with two numeric witnesses: the quadratic germ of Einstein–Hilbert equals $\frac{1}{2}S_{FP}$ (measured constant 0.500002, random metrics), and the second-order term of isotropic Schwarzschild exactly cancels the self-energy of the first-order field (symbolic, with the nonzero $-3\varepsilon^2/r^4$ residual of the uncorrected linear theory as the null). Only all-orders uniqueness remains cited [Wald, 1986].
- **(R-Gauss) generalised.** Via Cramér's theorem the quadratic effective action extends to arbitrary finite-moment i.i.d. modes; non-Gaussianity changes only the coefficient ($I'' = 1/\mathrm{Var}(x^2)$, verified exactly), which the Newtonian matching absorbs into G. Renamed (R-corr): only the strongly-correlated regime remains.
- **(R-cont) reduced to (R-GH).** Band exactness (sampled harmonics are exact eigenvectors, no limit needed) plus the explicit eigenvalue rate $(1-\rho_k) = (3n^2-7)\theta^2/60 + O(\theta^4)$ from the closed-form dispersion give field-dynamics continuum control with explicit constants; only the Gromov–Hausdorff statement about the arena geometry remains with Paper XL.
- **(R-G) conditionally derived.** Under the prior structural claim that the muon sits at cascade shell $96 = 24 \times 4$ (Paper LII), $G = \hbar c/(m_\mu \varphi^{96})^2 = 6.67305 \times 10^{-11}$, within 1.9×10^{-4} of measurement.
- **SM bridge upgraded.** The two-arena mechanism's gauge *inventory* is derived: $SU(3) \times SU(2)$ with a distinguished clock circle $U(1)_{\text{clock}} \subset SU(2)$ ($\mathrm{Der}(\mathbb{O}) = \mathfrak{g}_2$, clock-unit stabiliser $= \mathfrak{su}(3)$ identified by computation; internal left $SU(2)$ from the frame split; tower stopped by sedenion zero divisors) — see Paper LVI and `docs/gauge-group-from-arenas.md`. All three SM factor types appear; the SM's *independent* hypercharge factor, chirality, representations, and couplings remain open (this supersedes §12's earlier phrasing).

14 Verification manifest

All claims are verified by `scripts/verify_gr_closure.py` (39 checks, all passing, each with a falsifiable null).

Sim	Verifies	Result
T1	FP uniqueness: 1-dim null space, ratios $(-\frac{1}{2}, 1, -1, \frac{1}{2})$, masses killed	exact
T2	$\dim\ker = 6 \supset$ 4-dim gauge orbit $\Rightarrow$ 2 polarisations	exact
T3a	$\delta S_{FP}/\delta h = G^{\text{lin}}$	$< 10^{-15}$
T3	grid solve: factor 2, isotropy, full gauge-unfixed equation	exact/3%
T4	ray deflection $4GM/(bc^2)$; scalar control half	5%/0.5%
T5–T6	exact tick-energy and local continuity on V_{600}	$< 10^{-10}$
T7	$\Theta_{\mu\nu}$: positivity, conservation $O(\delta^2)$, dust limit	exact
T8	chart shift = gauge direction, $O(\xi^2)$	ratio 4.00
T9	TT waves on the light cone only	exact
T10	second difference and eigenmode frequency, $O(\tau^2)$	ratio 4.00
T11	envelope $\to$ Schrödinger, error $O(\epsilon^2)$	ratio 3.97
T12	inertial = gravitating mass on one packet	2×10^{-4}
T13	Maxwell uniqueness, 2 photon polarisations	exact
T14	rank-2 tensoriality on V_{600}, π_r equivariance	$< 10^{-10}$
T15	Wishart quadratic dominance, $N^{-1/2}$ scaling	1.00001

15 Reproducibility

This manuscript lives at `papers/paper-liii/paper-liii.tex`. The derivation source is `docs/gr-closure-derivation.md`; the verification suite is `scripts/verify_gr_closure.py` (run from the repository root with `python3 scripts/verify_gr_closure.py`; requires only NumPy and the bundled `scripts/600cell_data.npz`). The residual-closure second pass is `docs/residual-closure-derivation.md` and `docs/gauge-group-from-arenas.md`, verified by `scripts/verify_residual_closure.py` (24/24; NumPy, SciPy, SymPy). Companion suites: `verify_rendering_layer.py` (21/21), `verify_gap_strengthening.py` (22/22), `verify_boundary_green_function.py` (Newtonian tier). The pre-closure state of the chain, including the documented failure of the direct construction, is preserved in `docs/kappa-derivati` §6.7 as the honest record.

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